Model Complexity and Error

As model complexity grows, training error keeps falling, but test error first falls and then rises again once the model begins to overfit the training data.

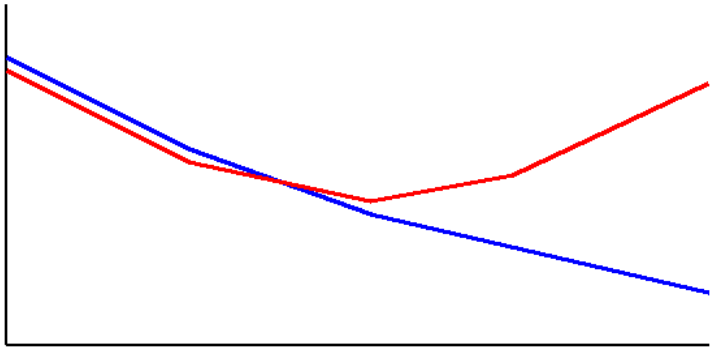

Figure 1: Training error and test error versus model complexity.

Choosing Complexity

The best model complexity is near the point where test error is lowest. A validation set is used to find that point without touching the test set.